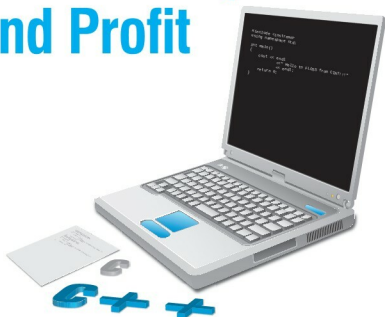


# C/C++ Code Interpretation for Fun and Profit



Since C and C++ are compiled languages, every time you make even a small change in your source code, you have to repeat the compile/link step before you can run it. While this is a trivial step for small programs, for even a medium-sized program, it is time-consuming and tedious. In contrast, scripting languages like Python, Ruby, Perl, etc. don't require compilation/linking, and you can even type code statements into interactive shells and obtain results. Such features for C/C++ code would be a great time-saver. As always, open source technologies come to the rescue! In this article, we explore various open source interpreters that run C/C++ programs directly, and accept code statements in an interactive shell. You will also learn how to harness the capabilities of some of these interpreters, beyond just C/C++ code interpretation.

C has been very popular for many decades because it's a performance-oriented high-level programming language with a lot of control over hardware resources. With its big brother C++, C is among the most-preferred high-level languages for software development even today. As we've noted, though, the time-consuming and boring compile-link-run cycle is annoying, especially to beginners just starting with C/C++ programming, and those who want to quickly prototype something in C/C++.

Scripting languages like Python, Ruby, Perl and others don't need code to be compiled

and linked, and you can even type code into their interactive shells for immediate results. Unfortunately, you can't switch to those when compatibility and performance are your primary goals; you have to stick with C and C++. The answer to this problem lies in the open source projects we will look at below. These run C/C++ programs directly, without any compile/link steps! Some also let you enter code statements in an interactive shell for immediate evaluation!

I used Puppy Linux 4.2.1 and the Ubuntu 9.10 64-bit desktop edition to test the C and C++ code used in this article and to generate various screen-shots.